



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2012
YEAR 11 Mathematics Extension
HSC Task #1

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answers must be given in simplest exact form unless otherwise stated.

Total marks - 77

Section I (10 marks)

- Answer Questions 1-10 on the Multiple Choice answer sheet provided.

Section II (67 marks)

- For Questions 11-13, start a new answer booklet for each question.

Examiner: *D.McQuillan*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

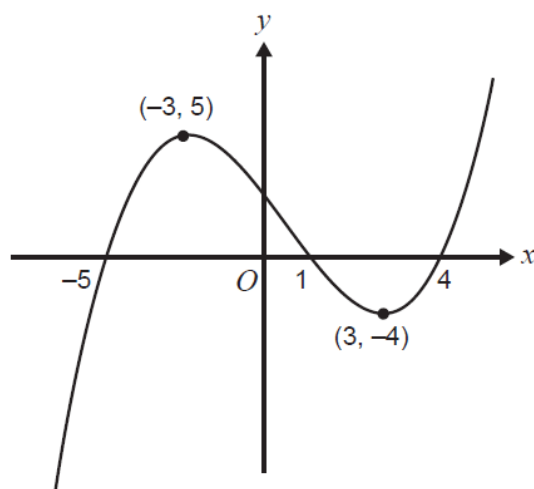
Section I (10 marks)

Use the multiple-choice answer sheet for Questions 1-10

- 1 What is the acute angle to the nearest degree between the lines $y = 7 - 4x$ and $2x - 3y - 6 = 0$?
- (A) 26°
- (B) 42°
- (C) 70°
- (D) 75°
- 2 When polynomial $P(x)$ is divided by $x^2 + x - 6$ the remainder is $7x + 13$. What is the remainder when $P(x)$ is divided by $x + 3$?
- (A) -8
- (B) -5
- (C) 34
- (D) 55
- 3 What is the exact value of $\cos 165^\circ$?
- (A) $\frac{-\sqrt{2}-\sqrt{6}}{4}$
- (B) $\frac{-\sqrt{2}+\sqrt{6}}{4}$
- (C) $\frac{-\sqrt{6}-\sqrt{2}}{4}$
- (D) $\frac{-\sqrt{6}+\sqrt{2}}{4}$

- 4 What are the coordinates of the point P that divides internally the interval joining the points A(1, 2) and B(7, 5) in the ratio 2:1 ?
- (A) (3, 3)
- (B) (3, 4)
- (C) (5, 3)
- (D) (5, 4)
- 5 What is the solution to the inequality $\frac{3}{x-2} \geq 4$?
- (A) $x < -2$ and $x \geq \frac{11}{4}$
- (B) $x > -2$ and $x \leq -\frac{11}{4}$
- (C) $x < 2$ and $x \geq \frac{11}{4}$
- (D) $x > 2$ and $x \leq \frac{11}{4}$
- 6 The average rate of change of the function $f(x) = x^3 - \sqrt{x+1}$ between $x = 0$ and $x = 3$ is
- (A) 12
- (B) $\frac{26}{3}$
- (C) $\frac{25}{3}$
- (D) 8

7



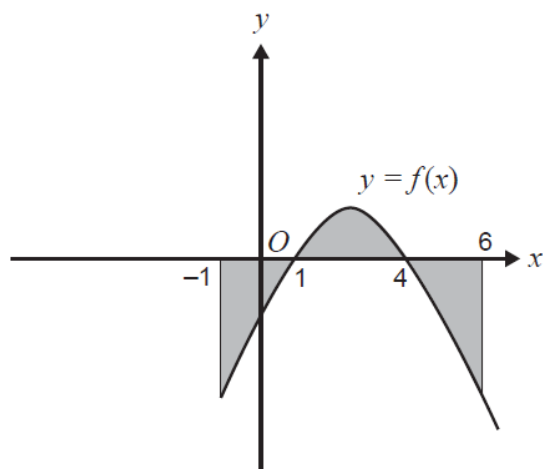
For the graph of $y = f(x)$ shown above $f'(x)$ is negative when

- (A) $-3 < x < 3$
- (B) $-3 \leq x \leq 3$
- (C) $x < -3$ or $x > 3$
- (D) $x < 5$ or $1 < x < 4$

8 Let $f'(x) = g'(x) + 3$, $f(0) = 2$ and $g(0) = 1$.

Then $f(x)$ is given by

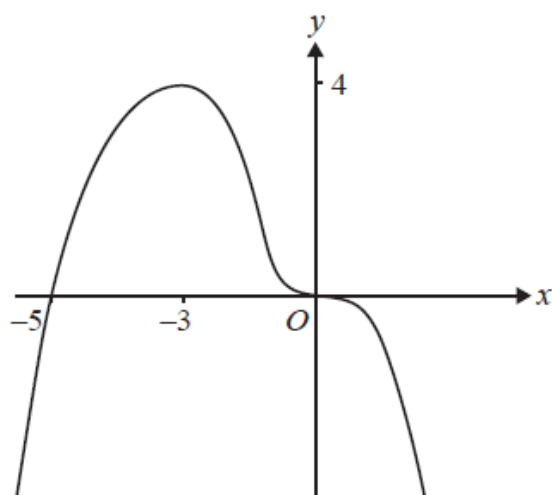
- (A) $f(x) = g(x) + 3x + 1$
- (B) $f(x) = g(x) + 3x$
- (C) $f(x) = 1$
- (D) $f(x) = g'(x) + 3$



The total area of the shaded regions in the diagram is given by

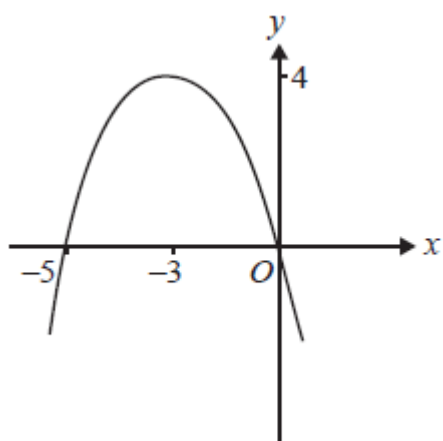
- (A) $\int_{-1}^6 f(x) dx$
- (B) $\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx + \int_4^6 f(x) dx$
- (C) $-\int_1^{-1} f(x) dx + \int_1^4 f(x) dx - \int_6^4 f(x) dx$
- (D) $\int_1^{-1} f(x) dx + \int_1^4 f(x) dx + \int_6^4 f(x) dx$

- 10 The graph of $y = f(x)$ is shown below.

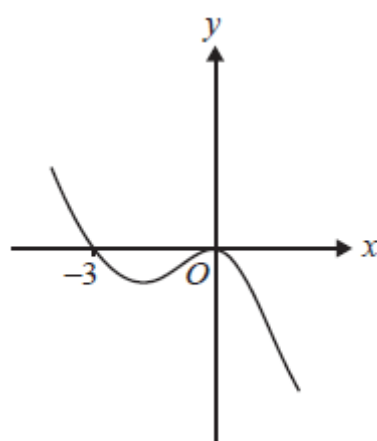


Which of the following could be the graph of the derivative function $y = f'(x)$?

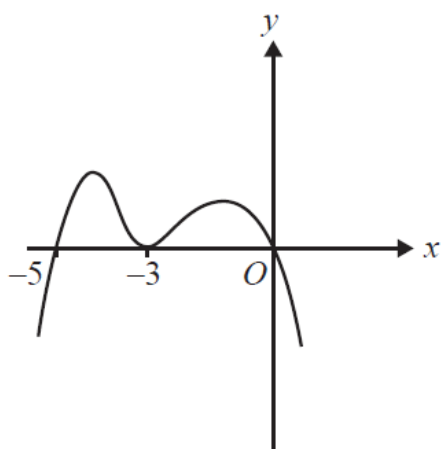
(A)



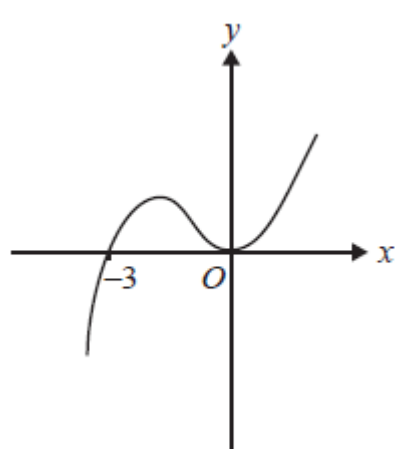
(B)



(C)



(D)



Section II

Start a new writing booklet for each Question in section II

Question 11 (21 marks)

(a) Find [4]

(i)

$$\int (5 - 2x + 6x^2) dx$$

(ii)

$$\int \sqrt[3]{x} \, dx$$

(iii)

$$\int \frac{5x^4 + x - 4}{2x^6} dx$$

(b) Solve [4]

(i)

$$\frac{4}{|x - 1|} > 1$$

(ii)

$$\frac{4}{(x - 1)^2} > 1$$

(c) Given $P(x) = x^3 - x^2 - 3x - 1$ [4]

(i) Evaluate $P(-1)$.

(ii) Hence solve $P(x) = 0$.

(d) The roots of $x^3 - 4x^2 - 9 = 0$ are α, β, γ . Find the value of [4]

(i) $(\alpha + \beta + \gamma) - \alpha\beta\gamma$

(ii) $(\alpha + 2)(\beta + 2)(\gamma + 2)$

(e) Prove that [2]

$$\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A$$

(f) Sketch the graph of a function that satisfies all of the given conditions. [3]

- $f'(5) = 0$
- $\lim_{x \rightarrow 3} f(x) = -\infty$
- $f'(x) < 0$ for $x < 3$ and for $x > 5$
- $f'(x) > 0$ for $3 < x < 5$

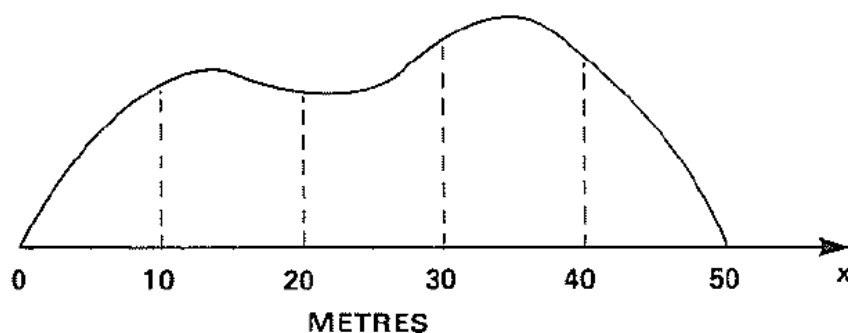
End of Question 11

Start a new writing booklet for each Question in section II

Question 12 (23 marks)

- (a) During a survey the area of an irregular headland was to be found. The surveyor used a base line divided into five equal sub-intervals, each of width ten metres. Offset measurements $f(x)$ were taken at each value x across the base line and tabulated as below

[3]



x (metres)	0	10	20	30	40	50
Offset (metres)	0	13.2	13.8	17	17.4	0

Use the trapezoidal rule to find the approximate area of the headland.

- (b) [4]

(i) Rewrite $2 \cos x + \sqrt{5} \sin x$ in the form $R \cos(x - \alpha)$.

(ii) Hence solve $2 \cos x + \sqrt{5} \sin x = 2$ for $0^\circ \leq x < 360^\circ$, to the nearest degree.

- (c) Sketch $y = x^4 - 6x^2 - 8x + 1$ showing all stationary points. [5]

- (d) If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ possibly be? [2]

(e) Find the general solution of $\cos 2x = \frac{1}{2}$ (answer in terms of radians). [2]

(f) [4]

(i) Show that

$$\frac{d}{dx} \left(\frac{x}{a^2 \sqrt{a^2 - x^2}} \right) = \frac{1}{\sqrt{(a^2 - x^2)^3}}$$

(ii) Hence evaluate

$$\int_0^2 \frac{dx}{\sqrt{(9 - x^2)^3}}$$

(g) Prove that the polynomial

$$P(x) = x^{101} + x^{51} + x + 1$$

has neither a local maximum nor a local minimum. [3]

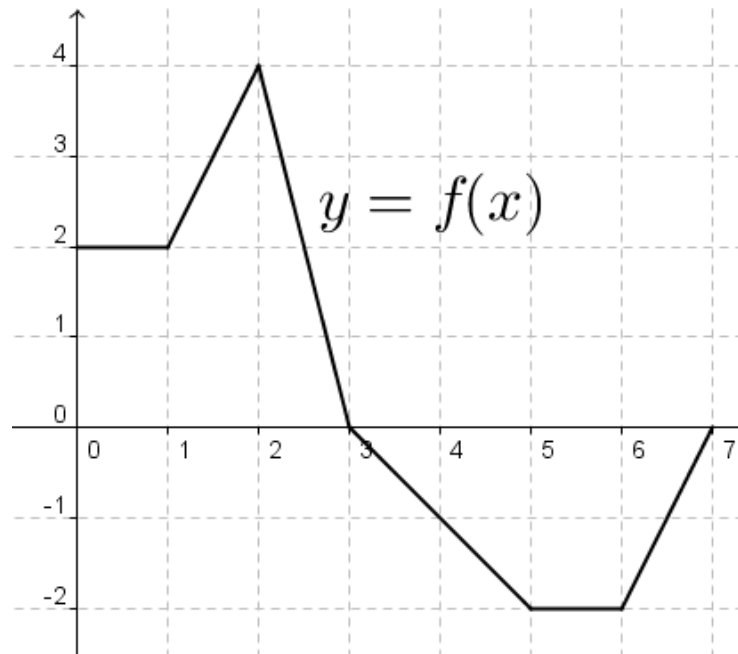
End of Question 12

Start a new writing booklet for each Question in section II

Question 13 (23marks)

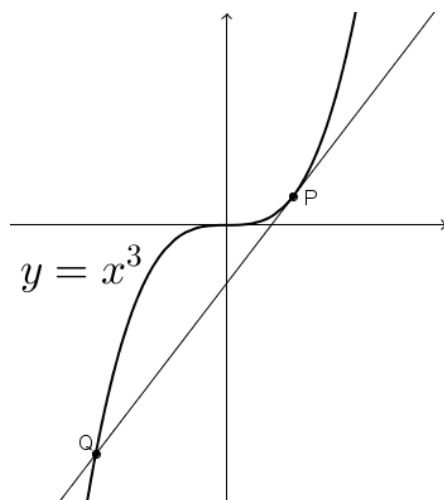
(a) Let $g(x) = \int_0^x f(t)dt$ where $f(x)$ is the function whose graph is shown.

[6]



- (i) Evaluate $g(1)$, $g(3)$ and $g(6)$.
- (ii) On what intervals is $g(x)$ increasing?
- (iii) Where does $g(x)$ have a maximum value?
- (iv) Sketch the graph of $y = g(x)$.

- (b) Let P be a point on the curve $y = x^3$ and suppose the tangent line at P intersects the curve again at Q. Prove that the slope at Q is four times the slope at P. [4]



- (c) A square room with length and breadth 4 metres is 2.4 metres high. Calculate, to the nearest minute, the angle which a string line will make with the floor if it is stretched from the centre of the ceiling to one corner of the floor. [3]

- (d) Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side L if one side of the rectangle lies on the base of the triangle. [4]

- (e) If $4 \sin 2x + 2 \cos 2x = \tan x$ [6]

(i) Prove that $\tan x$ is a root of the equation $t^3 + 2t^2 - 7t - 2 = 0$.

(ii) Verify that $t = 2$ is one root and find the exact values of the others.

(iii) Hence solve, $4 \sin 2x + 2 \cos 2x = \tan x$ for $0^\circ < x < 360^\circ$, to the nearest degree.

End of Question 13

End of Exam



Student Number: SOLNS

Mathematics Extension Task 1 2012

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. (A) (B) (C) (D)
2. (A) (B) (C) (D)
3. (A) (B) (C) (D)
4. (A) (B) (C) (D)
5. (A) (B) (C) (D)
6. (A) (B) (C) (D)
7. (A) (B) (C) (D)
8. (A) (B) (C) (D)
9. (A) (B) (C) (D)
10. (A) (B) (C) (D)

C
A
A or C
D
D
B
A
A
D
B

Q11 (EXT1)

(a) (i) $\int (5 - 2x + 6x^2) dx = 5x - x^2 + 2x^3 + C.$

[1]

(ii) $\int \sqrt[3]{x} dx = \int x^{\frac{1}{3}} dx.$

$$= \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + C.$$

$$= \frac{3}{4} x^{\frac{4}{3}} + C \quad \text{OR} \quad \frac{3x\sqrt[3]{x}}{4} + C.$$

[1]

(iii) $\int \frac{5x^4 + x - 4}{2x^6} dx = \int \left(\frac{5}{2} x^{-2} + \frac{x^{-5}}{2} - 2x^{-6} \right) dx$

$$= \frac{5x^{-1}}{-2} + \frac{x^{-4}}{-8} - \frac{2x^{-5}}{-5} + C$$

$$= \left[-\frac{5}{2} x^{-1} - \frac{1}{8} x^{-4} + \frac{2}{5} x^{-5} + C \right]$$

$$\text{OR} = \frac{-5}{2x} - \frac{1}{8x^4} + \frac{2}{5x^5} + C$$

$$\text{OR} \quad \left[\frac{-100x^4 - 5x + 16}{40x^5} \right] + C$$

[2]

(b) (i) $\frac{4}{|x-1|} > 1$

$$\Rightarrow |x-1| < 4$$

$$\text{ie } -4 < x-1 < 4$$

$$\text{ie } \boxed{-3 < x < 5} \quad \boxed{x \neq 1}$$

[2]

(ii) $\frac{4}{(x-1)^2} > 1$
 $4 > (x-1)^2$

$$-2 < x-1 < 2$$

$$\boxed{-1 < x < 3, x \neq 1}$$

[2]

$$(c) \quad P(x) = x^3 - x^2 - 3x - 1.$$

$$(i) \quad P(-1) = -1 - 1 + 3 - 1 \\ = 0.$$

$\therefore x+1$ is a factor.

$$-1 \left[\begin{array}{cccc} 1 & -1 & -3 & -1 \\ & -1 & 2 & 1 \\ \hline & 1 & -2 & -1 & 0 \end{array} \right] \quad \left[\begin{array}{l} \text{using Synthetic} \\ \text{or Long Division} \end{array} \right]$$

$\therefore x^2 - 2x - 1$ is the other factor.

$$\therefore P(x) = (x+1)(x^2 - 2x - 1)$$

$$\text{If } P(x) = 0$$

$$(x+1)(x^2 - 2x - 1) = 0$$

$$x = -1, \frac{2 \pm \sqrt{8}}{2}$$

$$\boxed{x = -1; 1 \pm \sqrt{2}.}$$

$$(d) \quad \text{given } x^3 - 4x^2 - 9 = 0 \text{ with roots } \alpha, \beta, \gamma.$$

$$\alpha + \beta + \gamma = 4$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0$$

$$\alpha\beta\gamma = 9.$$

$$\therefore (i) \quad (\alpha + \beta + \gamma) - \alpha\beta\gamma = 4 - 9 \\ \boxed{= -5}$$

$$(ii) \quad (\alpha+2)(\beta+2)(\gamma+2) = \alpha\beta\gamma + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ + 4(\alpha + \beta + \gamma) + 8. \\ = 9 + 0 + 16 + 8 \\ = \boxed{33}$$

(e) Prove that $\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A$.

$$\text{LHS} = \frac{1 - (1 - 2\sin^2 A)}{1 + (2\cos^2 A - 1)}$$

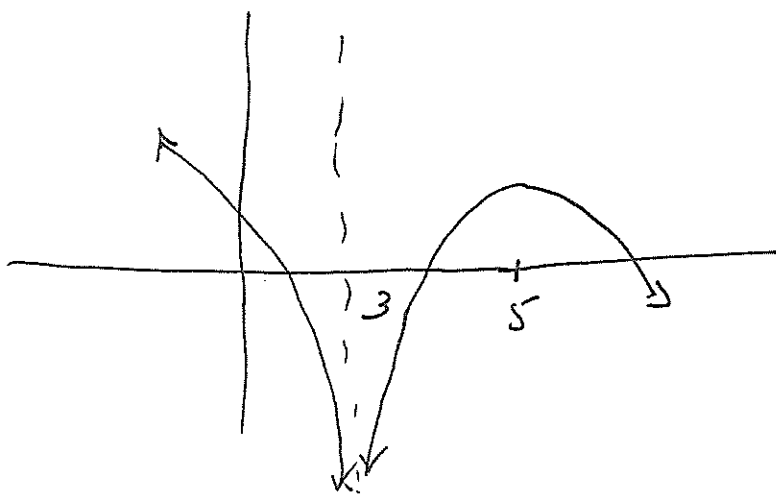
[2]

$$= \frac{2\sin^2 A}{2\cos^2 A}$$

$$= \tan^2 A$$

$$= \text{RHS.}$$

(f)



[3]

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Question 12

$$\begin{aligned} \text{(a)} \quad A &\doteq \frac{h}{2} [y_0 + y_5 + 2(y_1 + y_2 + y_3 + y_4)] \\ &= 5 [0 + 0 + 2(13.2 + 13.8 + 17 + 17.4)] \\ &= 614 \text{ m}^2 \quad [3] \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{(i)} \quad 2 \cos x + \sqrt{5} \sin x &= R \cos(x - a) \end{aligned}$$

$$R = \sqrt{2^2 + 5^2}$$

$$= 3$$

$$\therefore \text{LHS} = 3 \left(\frac{2}{3} \cos x + \frac{\sqrt{5}}{3} \sin x \right)$$

$$\therefore \sin a = \frac{\sqrt{5}}{3}$$

$$a \doteq 48^\circ 11' 23''$$

$$\doteq 0.8411^c \quad [2]$$

$$\begin{aligned} \text{(ii)} \quad E_{\text{op}} &= 3 \cos(x - 0.8411) \\ &= 3 \cos(x - 48^\circ 11' 23'') \end{aligned}$$

$$3 \cos(x - 48^\circ 11' 23'') = 2$$

$$x - 48^\circ 11' 23'' = \cos^{-1}\left(\frac{2}{3}\right)$$

$$= -48^\circ 11' 23'' \quad 48^\circ 11' 23''$$

$$x = 0^\circ \text{ or } 96^\circ 22' 46''$$

$$\doteq 0^\circ \text{ or } 96^\circ \quad [2]$$

$$\text{(c)} \quad y = x^4 - 6x^2 - 8x + 1$$

$$y' = 4x^3 - 12x - 8$$

$$y'' = 12x^2 - 12$$

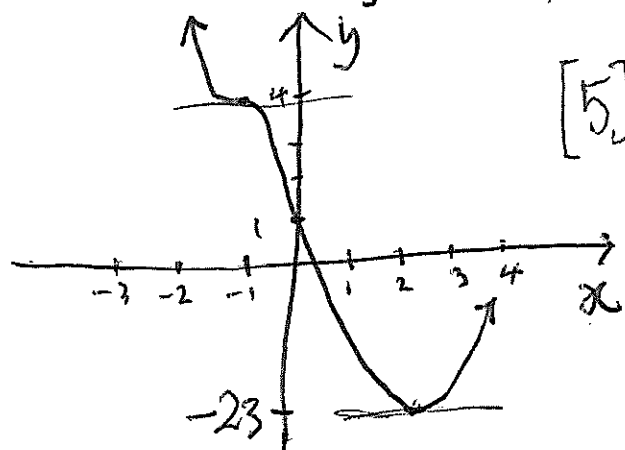
$$y' = 0 \text{ when}$$

$$4x^3 - 12x - 8 = 0$$

$$x^3 - 3x - 2 = 0$$

$$x = -1, -1, 2$$

$$y = 4, -23$$



(d) Min of $f(4)$ will be when $y'(x) = 2$ (least value)

$\therefore$ let $m = 2, (1, 10)$

$$y - 10 = 2(x - 1)$$

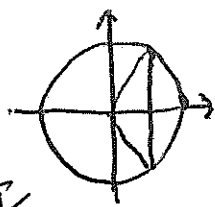
$$y = 2x + 8$$

When $x = 4$

$$y = 8 + 8 = 16$$

$$\therefore f(4) \geq 16 \quad [2]$$

$$(e) \cos 2x = \frac{1}{2}$$



$$2x = 2n\pi \pm \frac{\pi}{3}$$

$$x = n\pi \pm \frac{\pi}{6} \quad [2] \\ n \in \mathbb{Z}$$

$$(f) (i) \frac{d}{dx} \left(\frac{x}{a^2 \sqrt{a^2 - x^2}} \right)$$

$$= \frac{1}{a^2} \left[\frac{\sqrt{a^2 - x^2} \cdot 1 - x \cdot \frac{1}{2\sqrt{a^2 - x^2}} \cdot (-2x)}{(a^2 - x^2)} \right]$$

$$= \frac{1}{a^2} \left[\frac{(a^2 - x^2) + x^2}{\sqrt{(a^2 - x^2)^3}} \right]$$

$$= \frac{1}{\sqrt{(a^2 - x^2)^3}} \quad [2] \\ \text{as required.}$$

$$(ii) \int_0^2 \frac{dx}{\sqrt{(9 - x^2)^3}}$$

$$= \left[\frac{1}{9} \frac{x}{\sqrt{9 - x^2}} \right]_0^2$$

$$= \frac{1}{9} \left[\frac{2}{\sqrt{5}} - 0 \right]$$

$$= \frac{2}{9\sqrt{5}} \quad [2]$$

$$(g) P(x) = x^{101} + x^{51} + x + 1$$

$$P'(x) = 101x^{100} + 51x^{50} + 1$$

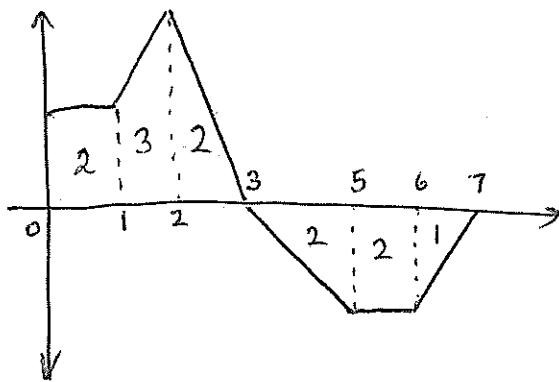
$$> 0 \quad \forall x$$

$\therefore P(x)$ is monotonic increasing.

$\therefore P(x)$ has no turning points. [3]

Question 13

a)



i) $g(1) = 2$

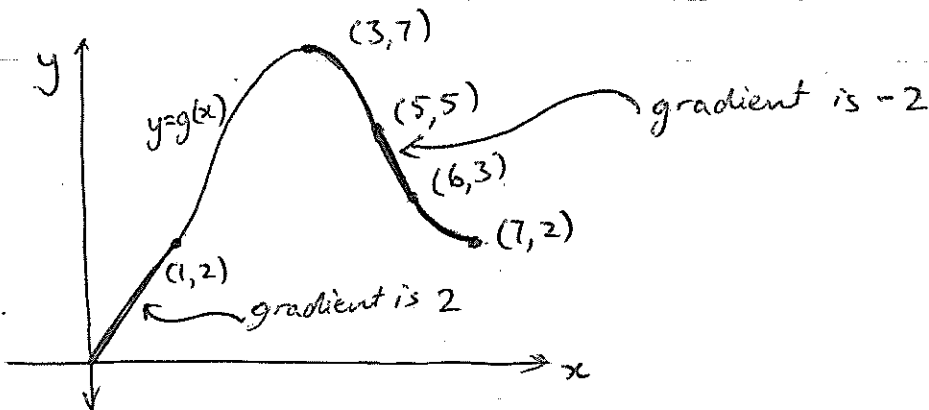
$g(3) = 7$

$g(6) = 3$

ii) $0 < x < 3$

iii) when $x = 3$

iv)



b) let P have coordinates (x_1, y_1)

Q have coordinates (x_2, y_2)

$$y = x^3$$

$$y' = 3x^2$$

at P

$$m_T = 3x_1^2$$

equation of tangent at P : $y - y_1 = 3x_1^2(x - x_1)$

$$y = 3x_1^2x - 3x_1^3 + y_1$$

$$y = 3x_1^2x - 2x_1^3 \quad \text{①, since } y_1 = x_1^3$$

$$y = x^3 \quad \text{②}$$

sub ① into ②

$$x^3 = 3x_1^2x - 2x_1^3$$

$$x^3 - 3x_1^2x + 2x_1^3 = 0$$

let roots be x_1, x_1, x_2 (since tangent at $P(x_1, y_1)$)

$$x_1 + x_1 + x_2 = 0$$

$$2x_1 + x_2 = 0$$

$$x_2 = -2x_1$$

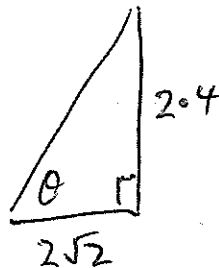
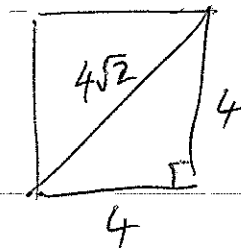
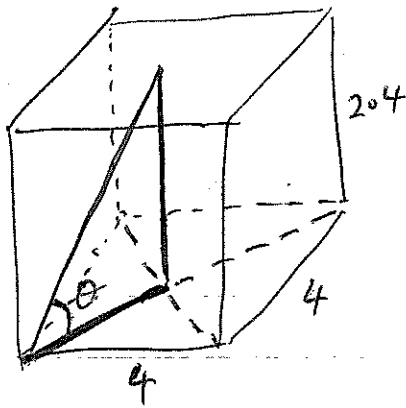
$$\text{gradient of tangent at } Q = 3(-2x_1)^2$$

$$= 12x_1^2$$

$$= 4(3x_1^2)$$

= four times gradient of tangent at P

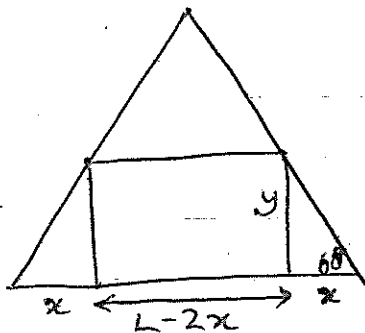
c)



$$\tan \theta = \frac{2 \cdot 4}{2\sqrt{2}}$$

$$\theta = 40^\circ 19'$$

d)



$$A = (L - 2x)y$$

$$\tan 60^\circ = \frac{y}{x}$$

$$\sqrt{3} = \frac{y}{x}$$

$$y = \sqrt{3}x$$

$$A = (L - 2x)\sqrt{3}x$$

$$A = \sqrt{3}Lx - 2\sqrt{3}x^2$$

$$A' = \sqrt{3}L - 4\sqrt{3}x$$

$$A'' = -4\sqrt{3}$$

$$\text{let } A' = 0$$

$$\sqrt{3}L - 4\sqrt{3}x = 0$$

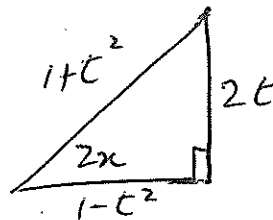
$$4\sqrt{3}x = \sqrt{3}L$$

$$x = \frac{L}{4}$$

since $A'' < 0$, Maximum area when $x = \frac{L}{4}$

Dimensions are $\frac{L}{2} \times \frac{\sqrt{3}L}{4}$

e) i) let $t = \tan x$



$$4 \sin 2x + 2 \cos 2x = \tan x$$

$$4 \left(\frac{2t}{1+t^2} \right) + 2 \left(\frac{1-t^2}{1+t^2} \right) = t$$

$$8t + 2 - 2t^2 = t + t^3$$

$$t^3 + 2t^2 - 7t - 2 = 0$$

ii) let $P(t) = t^3 + 2t^2 - 7t - 2$

$$P(2) = (2)^3 + 2(2)^2 - 7(2) - 2 = 0$$

$\therefore t = 2$ is a root of $P(t) = 0$.

$$\begin{array}{r} t^2 + 4t + 1 \\ t-2 \overline{) t^3 + 2t^2 - 7t - 2} \\ \underline{t^3 - 2t^2} \\ 4t^2 - 7t \\ \underline{4t^2 - 8t} \\ t - 2 \\ \underline{t - 2} \\ 0 \end{array}$$

$$\therefore P(t) = (t-2)(t^2 + 4t + 1)$$

$$t^2 + 4t + 1 = 0$$

$$t^2 + 4t + 4 = -1 + 4$$

$$(t+2)^2 = 3$$

$$t+2 = \pm\sqrt{3}$$

$t = -2 \pm \sqrt{3}$ are the other roots

iii) $\tan x = 2$

$\tan \alpha = 2$

$\alpha \doteq 63^\circ$

$x = 63^\circ, 243^\circ$

S	A ✓
✓ T	C

$\tan x = -2 + \sqrt{3}$

$\tan \alpha = 2 - \sqrt{3}$

$\alpha = 15^\circ$

$x = 165^\circ, 345^\circ$

✓ S	A
T	✓ C

$\tan x = -2 - \sqrt{3}$

$\tan \alpha = 2 + \sqrt{3}$

$\alpha = 75^\circ$

$x = 105^\circ, 285^\circ$

✓ S	A
T	✓ C

$\therefore x = 63^\circ, 105^\circ, 165^\circ, 243^\circ, 285^\circ, 345^\circ$